

Regex-Based Identification of Linguistic Patterns in Hindi Broadcast Speech

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Abstract

This paper proposes a rule-based system for identifying and analyzing seven linguistically significant patterns in Hindi broadcast speech using regular expressions (regex): reduplication, conjunctive verbs, compound verbs, honorifics, acronyms, multi-word expressions, and hyphenated word pairs. Patterns are extracted from three *Mann ki Baat* speech transcripts, first converted from Devanagari to ISO 15919 transliteration. Rule quality is evaluated using precision, recall, and F1 scores against a manually annotated gold standard. In the longer term, the discovered patterns will serve as features to identify dense sentences, enabling lightweight summarization. Hence, this work forms the foundation for an automatic summarization system that surfaces key points from a speech, so that readers do not need to process the full transcript to understand it.

1 Proposal

Problem Statement. Hindi is a morphologically rich, verb-final language whose surface forms encode tense, aspect, social register, and pragmatic stance through affixation, compounding, and particle use. Automatically identifying these structures is a prerequisite for downstream NLP tasks including information extraction, sentiment analysis, and automatic summarization of long-form speech. Most existing Hindi NLP tools rely on large annotated corpora or neural models that are computationally expensive and language-opaque (Agarwal et al., 2014) (Kakwani et al., 2020). This project asks: *can a set of hand-crafted regular expressions reliably identify seven linguistically motivated patterns in naturally occurring Hindi broadcast speech?*

Motivation. This project is driven by both technical and practical motivations:

- **Technical Efficiency:** Rule-based regex provides fast, transparent extraction without need-

ing annotated training data. Hindi’s predictable morphology (e.g., consistent stem-plus-suffix compound verbs) is highly suitable for this approach, allowing us to evaluate and refine the rules strictly using Precision, Recall, and F1-scores.

- **Practical Application:** Lengthy public broadcasts like *Mann ki Baat* are difficult for journalists and policymakers to parse quickly. By using these seven linguistic patterns as signals for informationally dense sentences, this system serves as the foundational step for an automated key-point extraction tool to summarize long Hindi speeches.

Proposed Approach. The system has two modules. A **transliteration module** maps raw Devanagari input to ISO 15919 romanisation. A **pattern extraction module** then applies seven rule sets, each evaluated independently by its precision, recall, and F1 score:

1. **Reduplication** — full word repetition signalling rhetorical emphasis or distribution (*dhīre-dhīre, ghar-ghar*);
2. **Conjunctive verbs** — verb stem + *-kar/-ke*, marking sequential or backgrounded actions (*dekh-kar, sun-ke*);
3. **Compound verbs** — verb stem followed by a vector verb (*lenā, denā, jānā, ānā, paṛnā, etc.*) conveying aspectual or pragmatic nuance (*kha lenā, bata denā*) (Chakrabarti et al., 2008);
4. **Honorifics** — the three-way pronoun register system: *tū* (intimate), *tuma* (informal), *āpa* (formal);
5. **Acronyms** — two or more consecutive uppercase Latin letters capturing institutional vocabulary;

6. **Multiword expressions (MWEs)** — fixed or semi-fixed multiword postpositions and formulaic phrases (*ke bād, kī vajaha se*) identified via lexicon-driven phrase patterns;
7. **Hyphenated word pairs** — two tokens joined by a hyphen covering compounds, echo words, and code-switched pairs (*Zero-defect, Fit-India, śuruāt-se*).

Per-pattern F1 scores are computed against the annotated gold standard.

2 Dataset

Source. The corpus consists of three specific transcripts of the *Mann ki Baat* broadcast dated December 2025, January 2026, and February 2026, sourced from the official PMO India archive ([Prime Minister’s Office, India, 2025–2026](#)).

Description. The 15,000-word corpus of broadcast Hindi spans diverse domains like governance, science, and cultural heritage. All target patterns occur naturally: compound verbs elaborate conversation, reduplication adds rhetorical stress, honorifics index formal address, and conjunctive verbs link narratives. This density and variety provide a robust dataset for computing meaningful F1 scores.

Relevance. *Mann ki Baat* occupies a unique cross-register position, blending formal Sanskrit-origin vocabulary with conversational compound verbs and English borrowings. This **mix of classical and contemporary speech** makes it an ideal stress test for our target regex rules. Beyond its linguistic value, the practical goal of this project is to use these extracted patterns as signals to automatically summarize key policy points and social initiatives from these thousand-word transcripts.

Preprocessing. Preprocessing has three stages: (1) UTF-8 validation and normalisation of Hindi punctuation, replacing the *poorna viram* with a sentence boundary marker; (2) ISO 15919 transliteration, mapping Devanagari characters to their diacritic Latin equivalents; and (3) whitespace normalisation. Latin-script tokens already in the source text are preserved for acronym, hyphenated pair, and borrowing detection.

Availability. All transcripts are freely and publicly available without registration. No proprietary data is used.

3 Deliverables

The following outputs will be produced by the end of the project:

- **Code:** A documented Python script containing the ISO 15919 transliteration pipeline and all seven regex rule sets, with per-pattern extraction and frequency counting over the full corpus.
- **Annotated evaluation set:** A manually labelled subset of 300 sentences drawn across all three transcripts, annotated for the presence or absence of each pattern, used as the gold standard for computing F1.
- **Evaluation results:** Precision, recall, and F1 scores for each of the seven patterns in a summary table, alongside per-episode frequency counts. F1 is used throughout as the primary measure of rule quality, balancing the cost of missed instances against false positives.
- **Written analysis:** A report containing (a) a cross-speech frequency comparison per pattern, (b) an error analysis categorising F1 shortfalls by cause — boundary ambiguity, transliteration conflict, or rule over-generalisation — and (c) a linguistic discussion of what the pattern distributions reveal about register and the feasibility of F1-guided key-point extraction.

4 Proposed Timeline

Week	Milestone
Week 1	Finalize scope; collect and explore all three transcripts; verify UTF-8 encoding and sentence boundaries.
Week 2	Build and validate the ISO 15919 transliteration pipeline; normalize punctuation; check output on sample sentences.
Week 3	Implement rules for reduplication, acronyms, and hyphenated word pairs; run on full corpus; begin literature review on Hindi morphology.
Week 4	Implement rules for conjunctive verbs, compound verbs, honorifics, and MWEs; manually annotate 300-sentence evaluation set.
Week 5	Compute precision, recall, and F1 for all seven patterns; conduct error analysis; refine underperforming rules.
Week 6	Write final report; compile code and annotations; submit PDF by the deadline.

Table 1: Project schedule.

References

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